

Inspiration Private School

Course Outline

Department: Science

MINISTRY OF EDUCATION

COURSE TITLE: Chemistry

GRADE LEVEL: 11

COURSE TYPE: University Preparation

COURSE CODE: SCH3U

DEVELOPED BY: Inspiration Private School Science Department, Ling Li, June 2017

DEVELOPED FROM:

Ministry of Education: Science, The Ontario Curriculum, Grades 11 and 12, 2008 (Revised)

REVISION DATE: August 2021

TEXT(S): *McGraw-Hill Ryerson Chemistry 11*

- *Webb et al; McGraw-Hill Ryerson, 2002*
- *Nelson Chemistry 12; 2003*
- Access to third party software for the online labs

PREREQUISITE: Grade 10 Science, Academic SNC2D

CREDIT VALUE: 1.0

LENGTH: 110 hours

COURSE DESCRIPTION:

This course enables students to deepen their understanding of chemistry through the study of the properties of chemicals and chemical bonds; chemical reactions and quantitative relationships in those reactions; solutions and solubility; and atmospheric chemistry and the behaviour of gases. Students will further develop their analytical skills and investigate the qualitative and quantitative properties of matter, as well as the impact of some common chemical reactions on society and the environment.

OUTLINE OF COURSE CONTENT:

Unit	Titles and Descriptions	Time and Sequence
Unit 1	Matter, Chemical Trends, and Chemical Bonding Students will develop an understanding of the structure of an element that allows us to predict its physical and chemical properties. They will investigate the physical and chemical properties of compounds based on the type of chemical bond formed. The appropriate use of chemicals will be examined in order to minimize the risks to human health and the environment.	21 hours
Unit 2	Chemical Reactions Students will demonstrate an understanding that chemicals react in predictable ways and will investigate different types of chemical reactions. They will assess the significant implications of chemical reactions and their applications for society and the environment.	22 hours
Unit 3	Quantities in Chemical Reactions Students will be introduced to the mole concept and how it can be used to describe chemical reactions quantitatively. They will also apply an understanding of quantitative relationships in determining the efficiency of chemical reactions and how such reactions can be optimized.	21 hours
Unit 4	Solutions and Solubility In this unit, students will demonstrate an understanding that properties of solutions can be described qualitatively and quantitatively, and can be predicted. They will assess the unique physical and chemical properties of water in supporting living organisms. Students will also analyse the origins of water pollution along with a variety of economic, social, and environmental issues related to drinking water.	23 hours

Unit 5	Gases and Atmospheric Chemistry Students will demonstrate an understanding of the concept of dynamic equilibrium and the variables that cause shifts in the equilibrium of chemical systems. They will investigate the qualitative and quantitative nature of chemical systems at equilibrium, and solve related problems. They will analyse chemical equilibrium processes, and assess their impact on biological, biochemical and technological systems.	21 hours
	Final Evaluation A two hour final exam, culminating activity, worth 30% of the final grade, meant as a summative evaluation of all strands, will be administered at the end of the course.	2 hours
	Total	110 hours

TEACHING AND LEARNING STRATEGIES:

As in a conventional classroom, instructors employ a range of strategies for teaching a course:

- Well-presented, clear writing and helpful graphics and diagrams
- Hands-on laboratory activities
- Research assignments, with direct instruction and coaching

In addition, teachers and students have at their disposal a number of tools that are unique to electronic learning environments:

- Electronic simulation activities
- Discussion boards and email
- Assessments with real-time feedback
- Interactive activities that engage both the student and teacher in subject

STRATEGIES FOR ASSESSMENT AND EVALUATION OF STUDENT PERFORMANCE:

Assessment is a systematic process of collecting information or evidence about a student's progress towards meeting the learning expectations. Assessment is embedded in the instructional

activities throughout a unit. The expectations for the assessment tasks are clearly articulated and the learning activity is planned to make that demonstration possible. This process of beginning with the end in mind helps to keep focus on the expectations of the course. The purpose of assessment is to gather the data or evidence and to provide meaningful feedback to the student about how to improve or sustain the performance in the course. Scaled criteria designed as rubrics are often used to help the student to recognize their level of achievement and to provide guidance on how to achieve the next level. Although assessment information can be gathered from a number of sources (the student, the student's course mates, the teacher), evaluation is the responsibility of only the teacher. Evaluation is the process of making a judgment about the assessment information and determining the percentage grade or level.

Assessment is embedded within the instructional process throughout each unit rather than being an isolated event at the end. Often, the learning and assessment tasks are the same, with formative assessment provided throughout the activity. In every case, the desired demonstration of learning is articulated clearly and the learning activity is planned to make that demonstration possible. This process of beginning with the end in mind helps to keep focus on the expectations of the course as stated in the course guideline. The evaluations are expressed as a percentage based upon the levels of achievement.

All curriculum expectations are accounted for in instruction and assessment. Evidence of student achievement for assessment is collected over time from three different sources: observations, conversations, and student products.

Learning Skills and Work Habits

The development, assessment, and evaluation of learning skills and work habits will occur in the following areas: responsibility, organization, independent work, collaboration, initiative, and self-regulation. Learning skills and work habits, apart from when learning skills and work habits may be included as part of a curriculum expectation, will not be considered in the determination of a student's grade.

Categories of Knowledge and Skills; Levels of Achievement

Students will have numerous and varied opportunities to demonstrate the full extent of their achievement of the content standards (curriculum expectations) and performance standards (achievement chart) in a balanced manner across all four categories of knowledge and skills: knowledge and understanding, thinking, communication, and application. Four levels of achievement (levels 1-4) are identified on the achievement chart for Science. Depending on the task or activity, students may receive marks, achievement levels, or comments.

Assessment for/as Learning

Assessment is the process of gathering information from a variety of sources that accurately reflects how well a student is achieving both specific and overall curriculum expectations in a course. As part of assessment, teachers and peers will provide students with descriptive feedback that guides their efforts towards improvement (assessment for learning) as well as guides them in assessing their own progress (assessment as learning).

Assessment of Learning

Evaluation is based on assessment of learning and focuses on students' achievement of the overall expectations. A single assessment of learning may include one or more of the four knowledge and skills categories. Assessment of learning in this course will be continuous throughout the year, will follow opportunities for students to improve their understanding, and will include a variety of assessment methods. Assessment of each student's learning is done independently and by the teacher.

Strategies for Assessment of Student Performance

Teachers of this course will work collaboratively with colleagues to regularly review course content, instructional strategies and assessment practices, and will make modifications to the program as needed to improve student achievement. Results from the AP, PSAT, SAT examinations, and province-wide assessments (OSSLT) will provide additional information on program effectiveness to the school.

Assessment or Evaluation Strategy (Term = 70%)	Purpose? As, For, Of Learning	Who is completing the assessment or evaluation?	Scoring Tool
Laboratory work (prepared or student-designed)	As Learning For Learning Of Learning	Peers Self Teacher	Rubric or marking scheme
Unit Tests	Of Learning	Teacher	Marking Scheme
Concept Tests	For Learning	Teacher	Marking Scheme
Presentations	As Learning For Learning Of Learning	Peers Self Teacher	Rubric or marking scheme
Homework	As Learning For Learning	Self/Peers Teacher	Marking scheme
Assessment or Evaluation Strategy (Summative = 30%)	Portion of the Course (Summative Work - 30%)	Who is completing the assessment or evaluation?	Scoring Tool
Final Summative Project	As Learning For Learning Of Learning	Peers Self Teacher	Rubric

Assessment or Evaluation Strategy (Term = 70%)	Purpose? As, For, Of Learning	Who is completing the assessment or evaluation?	Scoring Tool
Final Examinations (Written & Practical)	Of Learning	teacher	Marking Scheme

The teacher will ensure that student work is assessed and/or evaluated in a balanced manner with respect to the four categories (see below), and that achievement of particular expectations is considered within the appropriate categories. The four categories should be considered as interrelated, reflecting the wholeness and interconnectedness of learning. The four categories of knowledge and skills are: Knowledge and Understanding; Thinking; Communication; Application.

Category	Weight
Knowledge / Understanding	25%
Thinking / Inquiry	25%
Communication	25%
Application	25%

OVERALL CURRICULUM EXPECTATIONS:

Scientific Investigation Skills and Career Exploration	
Overall Expectations	
150.040.01.01	demonstrate scientific investigation skills (related to both inquiry and research) in the four areas of skills (initiating and planning, performing and recording, analysing and interpreting, and communicating);
150.040.01.02	identify and describe a variety of careers related to the fields of science under study, and identify scientists, including Canadians, who have made contributions to those fields.
Matter, Chemical Trends, and Chemical Bonding	
Overall Expectations	
150.040.02.01	analyse the properties of commonly used chemical substances and their effects on human health and the environment, and propose ways

	to lessen their impact;
150.040.02.02	investigate physical and chemical properties of elements and compounds, and use various methods to visually represent them;
150.040.02.03	demonstrate an understanding of periodic trends in the periodic table and how elements combine to form chemical bonds.
Chemical Reactions	
Overall Expectations	
150.040.03.01	analyse chemical reactions used in a variety of applications, and assess their impact on society and the environment;
150.040.03.02	investigate different types of chemical reactions;
150.040.03.03	demonstrate an understanding of the different types of chemical reactions.
Quantities in Chemical Reactions	
Overall Expectations	
150.040.04.01	analyse processes in the home, the workplace, and the environmental sector that use chemical quantities and calculations, and assess the importance of quantitative accuracy in industrial chemical processes;
150.040.04.02	investigate quantitative relationships in chemical reactions, and solve related problems;
150.040.04.03	demonstrate an understanding of the mole concept and its significance to the quantitative analysis of chemical reactions.
Solutions and Solubility	
Overall Expectations	
150.040.05.01	analyse the origins and effects of water pollution, and a variety of economic, social, and environmental issues related to drinking water;
150.040.05.02	investigate qualitative and quantitative properties of solutions, and solve related problems;
150.040.05.03	demonstrate an understanding of qualitative and quantitative properties of solutions.
Gases and Atmospheric Chemistry	
Overall Expectations	

150.040.05.01	analyse the cumulative effects of human activities and technologies on air quality, and describe some Canadian initiatives to reduce air pollution, including ways to reduce their own carbon footprint;
150.040.05.02	investigate gas laws that explain the behaviour of gases, and solve related problems;
150.040.05.03	demonstrate an understanding of the laws that explain the behaviour of gases.

The Final Grade:

The evaluation for this course is based on the student's achievement of curriculum expectations and the demonstrated skills required for effective learning. The percentage grade represents the quality of the student's overall achievement of the expectations for the course and reflects the corresponding level of achievement as described in the achievement chart for the discipline. A credit is granted and recorded for this course if the student's grade is 50% or higher. The final grade for this course will be determined as follows:

- 70% of the grade will be based upon evaluations conducted throughout the course. This portion of the grade will reflect the student's most consistent level of achievement throughout the course, although special consideration will be given to more recent evidence of achievement.
- 30% of the grade will be based on a final assessment task that occurs at or near the end of the course. In the case of this course, this final assessment task will take the form of a final examination which will be completed online under the supervision of a pre-approved proctor. The exam is worth 30% of the student's final mark in the course.

The Report Card:

The report card will focus on two distinct but related aspects of student achievement; the achievement of curriculum expectations and the development of learning skills. The report card will contain separate sections for the reporting of these two aspects.

A Summary Description of Achievement in Each Percentage Grade Range and Corresponding Level of Achievement		
Percentage Grade Range	Achievement Level	Summary Description
80-100%	Level 4	A very high to outstanding level of achievement. Achievement is <i>above</i> the provincial standard.
70-79%	Level 3	A high level of achievement. Achievement is <i>at</i> the provincial standard.
60-69%	Level 2	A moderate level of achievement. Achievement is <i>below</i> ,

		<i>but approaching</i> , the provincial standard.
50-59%	Level 1	A passable level of achievement. Achievement is <i>below</i> the provincial standard.
below 50%	Level R	Insufficient achievement of curriculum expectations. A credit will not be granted.

Achievement Chart: Science, Grades 9-12

Categories	50-59% (Level 1)	60-69% (Level 2)	70-79% (Level 3)	80-100% (Level 4)
Knowledge and Understanding - Subject-specific content acquired in each course (knowledge), and the comprehension of its meaning and significance (understanding)				
	The student:			
Knowledge of content (e.g., facts, terminology, definitions, safe use of equipment and materials)	demonstrates limited knowledge of content	demonstrates some knowledge of content	demonstrates considerable knowledge of content	demonstrates thorough knowledge of content
Understanding of content (e.g., concepts, ideas, theories, principles, procedures, processes)	demonstrates limited understanding of content	demonstrates some understanding of content	demonstrates considerable understanding of content	demonstrates thorough understanding of content
Thinking and Investigation - The use of critical and creative thinking skills and inquiry, research, and problem-solving skills and/or processes				
	The student:			
Use of initiating and planning skills and strategies (e.g., formulating questions, identifying the problem, developing hypotheses, selecting strategies and resources, developing plans)	uses initiating and planning skills and strategies with limited effectiveness	uses initiating and planning skills and strategies with some effectiveness	uses initiating and planning skills and strategies with considerable effectiveness	uses initiating and planning skills and strategies with a high degree of effectiveness
Use of processing skills and strategies (e.g., performing and recording, gathering evidence and data, observing, manipulating materials and using equipment safely, solving	uses processing skills and strategies with limited competence	uses processing skills and strategies with moderate competence	uses processing skills and strategies with considerable competence	uses processing skills and strategies with a high degree of competence

<i>equations, proving)</i>				
Use of critical / creative thinking processes, skills, and strategies (e.g., <i>analysing, interpreting, problem solving, evaluating, forming and justifying conclusions on the basis of evidence</i>)	uses critical / creative thinking processes, skills, and strategies with limited effectiveness	uses critical / creative thinking processes, skills, and strategies with some effectiveness	uses critical / creative thinking processes, skills, and strategies with considerable effectiveness	use critical / creative thinking processes, skills, and strategies with a high degree of effectiveness
Communication - The conveying of meaning through various forms				
	The student:			
Expression and organization of ideas and information (e.g., <i>clear expression, logical organization</i>) in oral, visual, and/or written forms (e.g., <i>diagrams, models</i>)	expresses and organizes ideas and information with limited effectiveness	expresses and organizes ideas and information with some effectiveness	expresses and organizes ideas and information with considerable effectiveness	expresses and organizes ideas and information with a high degree of effectiveness
Communication for different audiences (e.g., <i>peers, adults</i>) and purposes (e.g., <i>to inform, to persuade</i>) in oral, visual, and/or written forms	communicates for different audiences and purposes with limited effectiveness	communicates for different audiences and purposes with some effectiveness	communicates for different audiences and purposes with a high degree of effectiveness	communicates for different audiences and purposes with considerable effectiveness
Use of conventions, vocabulary, and terminology of the discipline in oral, visual, and/or written forms (e.g., <i>symbols, formulae, scientific notation, SI units</i>)	uses conventions, vocabulary, and terminology of the discipline with limited effectiveness	uses conventions, vocabulary, and terminology of the discipline with some effectiveness	uses conventions, vocabulary, and terminology of the discipline with considerable effectiveness	uses conventions, vocabulary, and terminology of the discipline with a high degree of effectiveness
Application - The use of knowledge and skills to make connections within and between various contexts				
	The student:			
Application of knowledge and skills (e.g., <i>concepts and processes, safe use of equipment, scientific investigation skills</i>) in familiar contexts	applies knowledge and skills in familiar contexts with limited effectiveness	applies knowledge and skills in familiar contexts with some effectiveness	applies knowledge and skills in familiar contexts with considerable effectiveness	applies knowledge and skills in familiar contexts with a high degree of effectiveness
Transfer of knowledge and skills (e.g., <i>concepts and</i>	transfers knowledge and	transfers knowledge and	transfers knowledge and	transfers knowledge and

<i>processes, safe use of equipment, scientific investigation skills) to unfamiliar contexts</i>	skills to unfamiliar contexts with limited effectiveness	skills to unfamiliar contexts with some effectiveness	skills to unfamiliar contexts with considerable effectiveness	skills to unfamiliar contexts with a high degree of effectiveness
Making connections between science, technology, society, and the environment (e.g., <i>assessing the impact of science on technology, people and other living things, and the environment</i>)	makes connections between science, technology, society, and the environment with limited effectiveness	makes connections between science, technology, society, and the environment with some effectiveness	makes connections between science, technology, society, and the environment with considerable effectiveness	makes connections between science, technology, society, and the environment with a high degree of effectiveness
Proposing courses of practical action to deal with problems relating to science, technology, society, and the environment	proposes courses of practical action of limited effectiveness	proposes courses of practical action of some effectiveness	proposes courses of practical action of considerable effectiveness	proposes highly effective courses of practical action

PROGRAM PLANNING CONSIDERATIONS FOR SCIENCE:

Teachers planning a program in Science must take into account considerations in a number of important areas. Essential information that pertains to all disciplines is provided in the companion piece to this document, *Science. The Ontario Curriculum, Grades 11 and 12: Some Considerations for Program Planning, 2008*. The areas of concern to all teachers include the following:

- Instructional Approaches
- Health and Safety in Science
- Planning Science Programs for Students with Special Education Needs
- Program Considerations for English Language Learners
- Environmental Education
- Antidiscrimination Education
- Critical Thinking and Critical Literacy in Science
- Literacy, Mathematical Literacy, and Investigation (Inquiry/Research) Skills
- The Role of Information and Communications Technology in Science
- The Ontario Skills Passport and Essential Skills
- Career Education
- Cooperative Education
- Planning Program Pathways and Programs Leading to a Specialist High Skills Major

Considerations relating to the areas listed above that have particular relevance for program planning in Science.

Instructional Approaches. Students bring to the classroom a natural curiosity as well as individual interests and abilities within their diverse personal and cultural experiences. Effective instructional approaches in Science draws upon their prior knowledge, captures their interest and encourages meaningful practice especially when the student sees a connection between what they are learning and their real-world application. Students are provided with opportunities to learn in a variety of ways. From a solid understanding of scientific concepts, the scientific method is employed to enable the student to investigate their world. The context for all learning in Science comes from the Relating Science to Technology, Society and the Environment (STSE) expectations.

Health and Safety in Science. The Science program provides the reading and analytical skills for the student to be able to explore the variety of concepts relating to health and safety in the workplace. Teachers who provide support for students in workplace learning placements need to assess placements for safety and ensure that students can read and understand the importance of issues relating to health and safety in the workplace. Teachers must also ensure that students have the knowledge and skills for safe participation in science activities.

Planning Science Programs for Students with Special Education Needs. Teachers plan programs that recognize the diversity of student strengths and needs by providing performance tasks that respect their student's particular abilities. The teacher will examine the strengths and learning needs of the student as well as the knowledge and skills that all students are expected to demonstrate in order to accommodate and/or modify the curriculum expectations as recorded in the student's Individual Education Plan (IEP).

Program Considerations for English Language Learners. Teachers will find positive ways to incorporate the diversity among the students into the online classroom environment. This Science course can provide a wide range of options to address the needs of ESL/ELD students. ESL programs are for students born in Canada or new comers whose first language is not English. ELD programs are for newcomers whose first language is not English. Assessment and evaluation exercises will help ESL students in mastering the English language and all of its idiosyncrasies. In addition, since all occupations require employees with a wide range of English skills and abilities, many students will learn how the operation of their own physical world can contribute to their success in their social world.

Environmental Education. The increased emphasis on relating science to technology, society, and the environment (STSE) provides numerous opportunities for teachers to integrate environmental education effectively into this science course.

Antidiscrimination Education. Antidiscrimination education promotes a school climate and classroom practice that encourages all students to work to high standards, ensures that they are given a variety of opportunities to be successful, affirms their self-worth, and helps them strengthen their sense of identity and positive self-image. The science curriculum is designed to help students acquire the habits of mind that are essential in a complex democratic society characterized by rapid technological, economic, political, and social change. These include respect and understanding with regard to individuals, groups, and cultures in Canada and the global community, including an appreciation and valuing of the contributions of Aboriginal people to the richness and diversity of Canadian life. The science curriculum also promotes respect and responsibility for the environment and an understanding of the rights, privileges, and responsibilities of citizenship. The learning activities in this science course should be inclusive in nature, reflecting diverse points of view and experiences. They should enable students to become more sensitive to the experiences and perceptions of others, to value and show

respect for diversity in the school and in the wider society, and to make responsible and equitable decisions in their own relationships. The critical thinking and scientific research skills acquired in science courses will enable students to recognize bias and stereotyping, as well as discriminatory attitudes that create barriers to productive relationships.

Critical Thinking and Critical Literacy in Science. Teachers plan science programs to promote critical thinking skills such as questioning, predicting, hypothesizing, analysing, synthesizing, examining opinions, identifying values and issues, detecting bias, and distinguishing between alternatives. As students work to achieve the STSE expectations, they are frequently asked to identify the implications of an action, activity, or process. In addition as students develop the skills of scientific investigation (inquiry/research skills), students are given the opportunity to ask appropriate questions to frame their research, interpret information, and detect bias. These learning activities would equip students with the skills to assess, analyze, and/or evaluate the impact of something on society and the environment.

Literacy, Mathematical Literacy, and Investigation (Inquiry/Research) Skills. Teachers plan science courses so that literacy, mathematical literacy, and investigation skills which are critical to students' success in all school subjects and in all areas of their lives, become integrated into the courses. To help students construct meaning from scientific texts, the science teachers model and teach the strategies that support learning to read while students are reading to learn in science. The science program builds on, reinforces, and enhances mathematical literacy. Students will also learn how to locate relevant information in a variety of print and electronic sources.

The Role of Information and Communications Technology in Science. Information technology is considered a learning tool that must be accessed by science students when the situation is appropriate. As a result, students will develop transferable skills through their experience with word processing, internet research, presentation software, and telecommunication tools, as would be expected in any environment.

The Ontario Skills Passport and Essential Skills. Teachers planning science programs are aware of the purpose and benefits of the Ontario Skills Passport (OSP). The OSP is a bilingual, web-based resource that enhances the relevance of classroom learning for students and strengthens school-work connections. The OSP provides clear descriptions of Essential Skills such as Reading Text, Writing, Computer Use, Measurement and Calculation, and Problem Solving and includes an extensive database of occupation-specific workplace tasks that illustrate how workers use these skills on the job. The Essential Skills are transferable, in that they are used in virtually all occupations. For further information on the OSP and the Essential Skills, visit [Ontario Skills Passport](#).

Career Education. Science definitely helps prepare students for employment in a huge number of diverse areas. The skills, knowledge and creativity that students acquire through this course are essential for a wide range of careers. Being able to express oneself in a clear concise manner without ambiguity, solve problems, make connections between this Science course and the larger world, etc., would be an overall intention of this Science course, as it helps students prepare for success in their working lives.

Cooperative Education. By applying the skills they have developed, students will readily connect their classroom learning to real-life activities in the world in which they live. Cooperative education and other workplace experiences will broaden their knowledge of employment opportunities in a wide range of fields. In addition, students will increase their understanding of workplace practices and the nature of

the employer-employee relationship. Teachers of Science should maintain links with community-based workers to ensure that students have access to hands-on experiences that will reinforce the knowledge they have gained in school.

Planning Program Pathways and Programs Leading to a Specialist High Skills Major. Teachers plan their science courses to be well suited for inclusion in some programs leading to a *Specialist High Skills Major (SHSM)* or in programs designed to provide pathways to particular apprenticeship or workplace destinations. Science courses may also be combined with cooperative education credits to provide the workplace experience required for some SHSM programs and for various program pathways to apprenticeship and workplace destinations.